Rajiv Gandhi University of Knowledge Technologies -RK Valley - AP

Academic Year 2024– 25: Semester 2 - Mid Semester Test-3

Computer Science & Engineering - Data strcures

R22Batch & Regular Mode of Examinations

Year: ENGG-1 | Date of Examination: 15-04- 2025

Time: 60 Minutes R220711 | Max Marks: 15

3X5 Marks= 15

The Question Paper Consists of 6 questions in 3 sets. Students need to answer 1 question from each set. Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

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1. Construct Binary tree , Binary Search tree and Min-heap for the following list: 4,2,1,9,12 5,6,7,3.

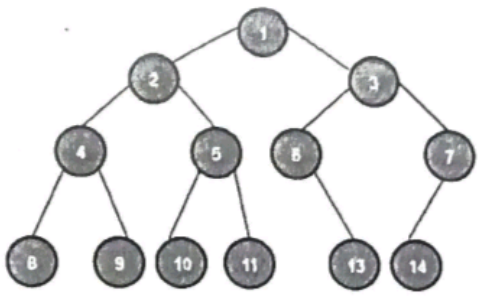
(OR)

2. Construct AVL tree for the following list: 6,4,10,8 ,1,3,2,5,9,7.

3. Construct B- trees for the following list "2,1,7,6, 5,3,4,8,10,9" (Note: M=3, Insert elements one by one).

(OR)

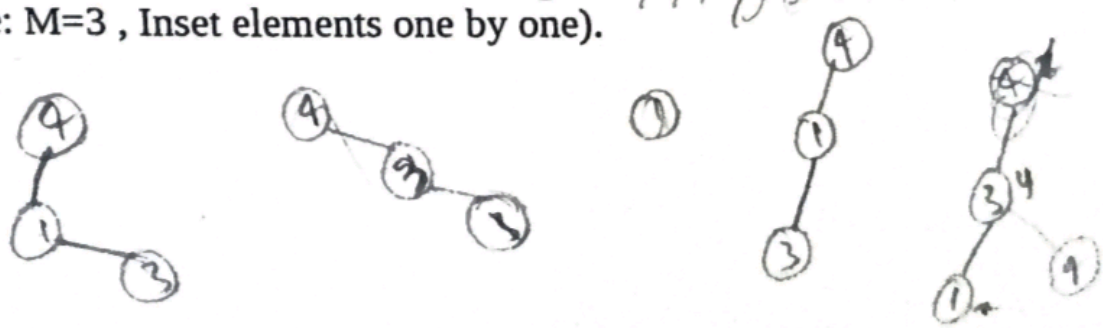
4. Find Pre-order, In-order and Post-order for the following tree.



5. Construct Red-Black tree for the following list "2,1,7,6, 5,3,4,8,10,9" (Note: Inset elements one by one).

(OR)

6. Construct B+ tree for the following list "2,1,7,6, 5,3,4,8,10,9" (Note: M=3 , Inset elements one by one).



Rajiv Gandhi University of Knowledge Technologies - RK Valley- AP

Academic Year 2024 – 25: Semester II - Mid Semester Test-3

Computer Science & Engineering - 23CS1201- Object Oriented Programming Through Java

R22 Batch & Regular Mode of Examinations

Year: ENGG-1 **Date of Examination: 16-April-2025**

Time: 60 Minutes R220711 **Max Marks: 15**

3X5 Marks= 15

The Question Paper consists of 3 units. Answer one full question from each unit.
Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

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1. Discuss how they are used to control access across packages. Provide examples demonstrating public, protected, default, and private access in package structures.

OR

2. Explain how to create and use user-defined packages with a complete Java program. Also, describe the advantages of using packages.

3. Explain inter-thread communication in Java. Discuss the use of wait(), notify(), and notifyAll() methods with an example code..

OR

4. Discuss the various ways to create a thread in Java. Compare and contrast using the Thread class and the Runnable interface with code examples.

5. Write a Java program using Swing to create a window with basic components. Include labels, text fields, and buttons arranged using a layout manager.

OR

6. Explain Action Listeners in Java Swing. Write a program to display a message when a button is clicked.

Rajiv Gandhi University of Knowledge Technologies - RK Valley- AP

Academic Year 2024-25: Semester II- Mid Semester Test-III

CSE - 20PY1201 – Engineering Physics

R22 Batch

Year: ENGG-1 Date of Examination: 16th April, 2025

Time: 60 Minutes R220711 Max Marks: 15

3X5 Marks= 15

The Question Paper consists of 3 units. Answer one full question from each unit.
Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

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1. Derive a three dimensional time independent Schrodinger Equation and write its two applications. [5 M]

OR

2. Calculate the eigenvalues and eigenfunction for a particle trapped in a 3-D square well potential. [5 M]

3. Define/describe/discuss [5 M]

a. Uncertainty principle b. Compton Effect

c. Photo electric effect d. De-Broglie matter wave

e. Wave function

OR

4. Define Fermi level. How does Fermi level depends on carrier-concentration and temperature, describe using equilibrium carrier statistics. [5 M]

5. What is Hall Effect and Hall Voltage? Derive the relation between Hall voltage and Hall coefficient. How Hall effect can be used for sensing? [5 M]

OR

6. A) Based on E-K curve how can we distinguish between semiconductors? [1.5 M]

B) How a PN Junction is formed because of diffusion? Explain with diagram, its construction, IV characteristic and energy band diagram. [2.5 M]